

Problem 1

Since the ambulance can only move vertical or horizontal, $X = |x| + |y|$ where x is the x choord and y is the y choord.

Now we can spilt our cdf into two parts the first part where X is less than or equal to 1. When you draw this on the grid this is when the diamond is touching or inside the grid thus $F_X(d) = (\sqrt{2}d)^2/4 = d^2/2$

The other is when $1 < X \leq 2$ because 2 is the max distance it can travel. This is when the diamond overlaps with the square thus we get $\frac{2d^2 - 4(d-1)^2}{4} = \frac{d^2 - 2(d-1)^2}{2}$

$$\text{Thus } F_X(d) = \begin{cases} \frac{d^2}{2} & 0 \leq d \leq 1 \\ \frac{d^2 - 2(d-1)^2}{2} & 1 < d \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

$$f_X(d) = F'_X(d) = \begin{cases} d & 0 \leq d \leq 1 \\ d - 2(d-1) & 1 < d \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

Problem 2

$$1. \quad (a) \quad \int_{-\infty}^{\infty} c(1-x^2) dx = 1$$

$$c \int_{-1}^1 (1-x^2) dx = 1$$

$$\left[x - \frac{x^3}{3} \right]_{-1}^1 = \frac{1}{c}$$

$$\left[1 - \frac{1}{3} + 1 - \frac{1}{3} \right] = \frac{1}{c}$$

$$\left[\frac{4}{3} \right] = \frac{1}{c}$$

$$c = \frac{3}{4}$$

$$F_X = \frac{3}{4} \int_{-1}^x (1-t^2) dt$$

$$= \frac{3}{4} \left(t - \frac{t^3}{3} \right)_{-1}^x = \frac{3}{4} \left(x - \frac{x^3}{3} + 1 - \frac{1}{3} \right)$$

$$(b) \quad \int_0^{\pi} c \sin(x) dx = 1$$

$$c [-\cos(x)]_0^{\pi} = 1$$

$$c [-\cos(\pi) + \cos(0)] = 1$$

$$c = \frac{1}{2}$$

$$F_X = \frac{1}{2} \int_0^x \sin(t) dt$$

$$= \frac{1}{2} (-\cos(t))_0^x = \frac{1}{2} (-\cos(x) + 1)$$

$$(c) \quad \int_0^{\infty} c(e^{-x} - e^{-2x}) dx = 1$$

$$c \lim_{t \rightarrow \infty} \left[-e^{-x} + \frac{1}{2} e^{-2x} \right]_0^t = 1$$

$$c \lim_{t \rightarrow \infty} \left[-e^{-t} + \frac{1}{2} e^{-2t} + \frac{1}{2} \right] = 1$$

$$\frac{c}{2} = 1$$

$$c = 2$$

$$F_X = 2 \int_0^x (e^{-t} - e^{-2t}) dt$$

$$= 2 \left(-e^{-t} + \frac{1}{2} e^{-2t} \right)_0^x$$

$$= 2 \left(-e^{-x} + \frac{1}{2} e^{-2x} + \frac{1}{2} \right)$$

$$2. \quad F_{X^2} = \mathbb{P}(X^2 \leq y) = \mathbb{P}(-\sqrt{y} \leq X \leq \sqrt{y}) = \int_{-\sqrt{y}}^{\sqrt{y}} f(x) dx$$

$$= \int_{-\sqrt{y}}^0 f(x) dx + \int_0^{\sqrt{y}} f(x) dx$$

$$F'_{X^2} = f_{X^2} = f(\sqrt{y}) \left(\frac{1}{2\sqrt{y}} \right) - f(-\sqrt{y}) \frac{-1}{2\sqrt{y}}$$

$$= \frac{1}{4\sqrt{y}} (e^{-|\sqrt{y}+1|} + e^{-|-\sqrt{y}+1|})$$

Problem 3

$$F_Z = \mathbb{P}(\max\{X_1, \dots, X_n \leq z\}) = \mathbb{P}(X_1 \leq z \text{ and } \dots \text{ and } X_n \leq z) = \mathbb{P}(X_1 \leq z) \cdots \mathbb{P}(X_n \leq z) = (F_{X_1})^n$$

$$F_{X_1} = \int_0^z 2x \, dx = z^2$$

$$F_Z = z^{2n}$$

$$F'_Z = f_Z = (2n)z^{2n-1} \text{ for } 0 \leq z \leq 1$$

$$E[Z] = \int_0^1 z(2n)z^{2n-1} \, dz = 2n \int_0^1 z^{2n} \, dz = 2n \left[\frac{z^{2n+1}}{2n+1} \right]_0^1 = \frac{2n}{2n+1}$$

Problem 4

$$1. \int_0^1 \int_0^{1-x} c e^{-(x+y)} dy dx = 1$$

$$c \int_0^1 \int_0^{1-x} e^{-(x+y)} dy dx = 1$$

Let $u = -y - x$ thus $du = -dy$

$$c \int_0^1 (-1) \int_x^1 e^u dy dx = 1$$

$$c \int_0^1 e^{-x} - e^{-1} dx = 1$$

$$c [-e^{-x} - e^{-1}x]_0^1 = c(-e^{-1} - e^{-1} + 1) = c(1 - 2e^{-1}) = 1$$

$$c = \frac{1}{1 - 2e^{-1}}$$

$$2. E[XY] = c \int_0^1 \int_0^{1-x} xye^{-(x+y)} dy dx = c \int_0^1 xe^{-x} \int_0^{1-x} ye^{-y} dy dx$$

$$= c \int_0^1 xe^{-x}((x-2)e^{x-1} + 1) dx = c(-xe^{-x} - e^{-x} + (e^{-1}/3)x^3 - e^{-1}x^2)_0^1$$

$$= c(-e^{-1} - e^{-1} + (e^{-1}/3) - e^{-1} + 1) = c(1 - (8e^{-1}/3))$$

Problem 5

Since X, Y, Z are independent and have the same pdf this implies they are identically distributed thus there are $3! = 6$ possible orderings with equal probabilities of X, Y, Z

Given $X > Y$ we have these orderings

$X > Y > Z, X > Y > Z$, and $Z > X > Y$

Thus $\mathbb{P}(X > Z \mid X > Y) = \frac{2}{3}$

Problem 6

1. Let $Z = X + Y$

Shown in class $f_Z(z) = \int_0^z (xe^{-x})(z-x)(e^{-z+x}) dx$ for $z \geq 0$

$$= e^z \int_0^z y(z-y) dy = (1/6)z^3 e^{-z} \text{ for } z \geq 0 \text{ and } 0 \text{ otherwise}$$

2. Let $Z = X + Y$ and $W = X/(X + Y)$ thus $X = WZ$ and $Y = Z - WZ$

Look at case $x, y \geq 0$ since the pdf would be zero otherwise

Thus shown in class $f_{W,Z}(w,z) = f_{X,Y}(wz, z - wz)(1/|\det(J)|)$

$$J = \begin{pmatrix} \frac{Y}{X+Y} & \frac{-X}{(X+Y)^2} \\ 1 & 1 \end{pmatrix} \text{ Thus } f_{W,Z}(w,z) = (z)(wze^{-wz})((1-w)ze^{-(z-wz)}) = w(1-w)z^3 e^{-z}$$

$$\text{Thus } f_W(w) = w(1-w) \lim_{t \rightarrow \infty} \int_0^t z^3 e^{-z} dz$$

$$= 6w(1-w) \text{ for } 0 \leq w \leq 1 \text{ and } 0 \text{ otherwise.}$$